Essential in STAAD.Pro (40 Hours)				
S.No	**Session**	**Topics Covered**	**Type**	**Duration (hrs)**
1	Session 1	Overview of Structural Analysis and Design Overview to STAAD Pro Connect Edition Overview of Fundamentals GUI Creating and opening a new file Unit System & Coordinate system Navigation & Mouse Controls STAAD Editor	Theory/Practical	2.0 Hrs
2	Session 2	Excercise Inclucded for practice	Practical	2.0 Hrs
3	Session 3	Grids & Types of Grids **Node Creation:** Snap Node, Node table, at overlapping beams, Insert Node for single and multiple members, using STAAD Editor. **Beam Creation:** Snap beam, using beam layout, connect beams along, Point to Point, Between Midpoints, Perpendicular Intersection, Curved Member and using STAAD Editor. **Structure Wizard:** Units, creating truss, creating frame, creating continuous beam, creating cylindrical frame and beam model, Editing parameters, Transferring the model to STAAD Pro, adding items to the library.	Theory/Practical	2.0 Hrs
4	Session 4	Excercise Inclucded for practice	Practical	2.0 Hrs
5	Session 5	Select Menu **Model Editing Tools for beam:** Split, stretch, break, Merge, Renumber and Move **Model Editing Tools for node:** Move, Merge & Renumber Translational Repeat Circular Repeat Mirror Rotate View Tools	Theory/Practical	2.0 Hrs
6	Session 6	Excercise Inclucded for practice	Practical	2.0 Hrs
7	Session 7	Material Specification Section Profile Specification Member Orientation **Member Specification:** Axial Actions, Member end release, Member Offset Support Specification Group Specification	Theory/Practical	2.0 Hrs
8	Session 8	Excercise Inclucded for practice	Practical	2.0 Hrs
9	Session 9	Creating a Primary Load, Adding Self weight Nodal Load **Member Load:** Uniform Force and Moment, Concentrated Force and Moment, Linear Varying Load, Trapezoidal Load, Hydrostatic Load, Pre/Post Stress Area Load Floor Load	Theory/Practical	2.0 Hrs
10	Session 10	Excercise Inclucded for practice	Practical	2.0 Hrs
11	Session 11	**Wind Load:** Defining Wind Load, Wind Load case **Load Combination:** Defining Load Combinations, Automatic Load Combination, Edit Auto Combination Rules Introduction to Analysis	Theory/Practical	2.0 Hrs
12	Session 12	Excercise Inclucded for practice	Practical	2.0 Hrs
13	Session 13	Introduction to FEM **Plate Generation:** 3 or 4-noded 2D plates, Surface objects with holes **Add Plate:** Triangular plate, Infill plates **Generate Meshing:** Plate Meshing, Surface Meshing **Solid Generation:** Solid 3D bricks from 4 to 8-noded Composite Decks **Structure Wizard:** Surface, Plate & Solid Models **Property Specification:** Thickness	Theory/Practical	2.0 Hrs
14	Session 14	Excercise Inclucded for practice	Practical	2.0 Hrs
15	Session 15	**Plate Load:** Pressure on Full Plate, Concentrated Load, Partial Plate Pressure Load, Trapezoidal Load, Hydrostatic Load Element Joint Load Moving Load	Theory/Demo	2.0 Hrs

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		Reference Load & Repeat Load		
16	Session 16	Excercise Inclucded for practice	Practical	2.0 Hrs
17	Session 17	Batch Design Vs Interactive Design **Concrete Design in STAAD.Pro:** Column Design, Beam Design	Theory/Practical	2.0 Hrs
18	Session 18	Excercise Inclucded for practice	Practical	2.0 Hrs
19	Session 19	Steel Design in STAAD.Pro Design of Overhead Transmission Line Towers. Report Generation	Theory/Practical	2.0 Hrs
20	Session 20	Excercise Inclucded for practice	Practical	2.0 Hrs